Total unimodularity

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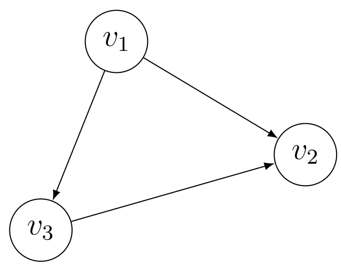
Solution of the practice quiz

The network is represented hereafter.



1. The incidence matrix has as many rows as nodes in the network, and as many columns as arcs in the network. In this case, it is a 3×3 matrix. We consider the arcs in the following order: (v_1, v_2), (v_1, v_3), (v_3, v_2). Each of them corresponds to a column that contains only two non zero entries: 1 at the row corresponding to the upstream node, and -1 at the row corresponding to the dowstream node. Therefore, the incidence matrix of this network is
$$A = \begin{pmatrix} 1 & 1 & 0 \\ -1 & 0 & -1 \\ 0 & -1 & 1 \end{pmatrix}.$$

2. Each square submatrix of size 1 is 0, 1 or -1. The determinant of each square submatrix of size 2 are

$$\det\begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix} = 1, \qquad \det\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} = -1, \qquad \det\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = 1,$$

$$\det\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = -1, \qquad \det\begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} = 1, \qquad \det\begin{pmatrix} 0 & -1 \\ -1 & 1 \end{pmatrix} = -1,$$

$$\det\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1, \qquad \det\begin{pmatrix} 1 & 0 \\ -1 & -1 \end{pmatrix} = -1, \qquad \det\begin{pmatrix} -1 & -1 \\ 0 & 1 \end{pmatrix} = -1.$$

Furthermore,

$$\det(A) = 0.$$

Then, the matrix is totally unimodular, as expected from the theory.